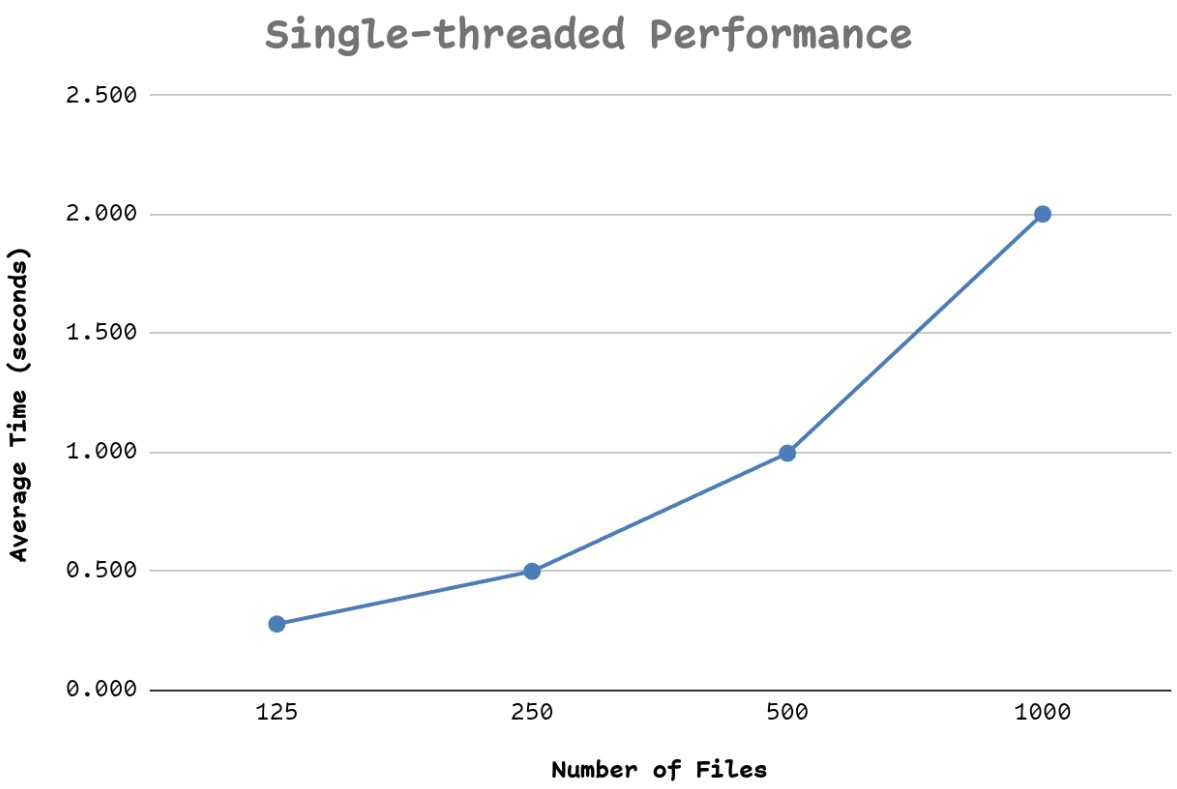
Full source code: https://github.com/lukedaoo/zod-threadscan

Experiment 1:



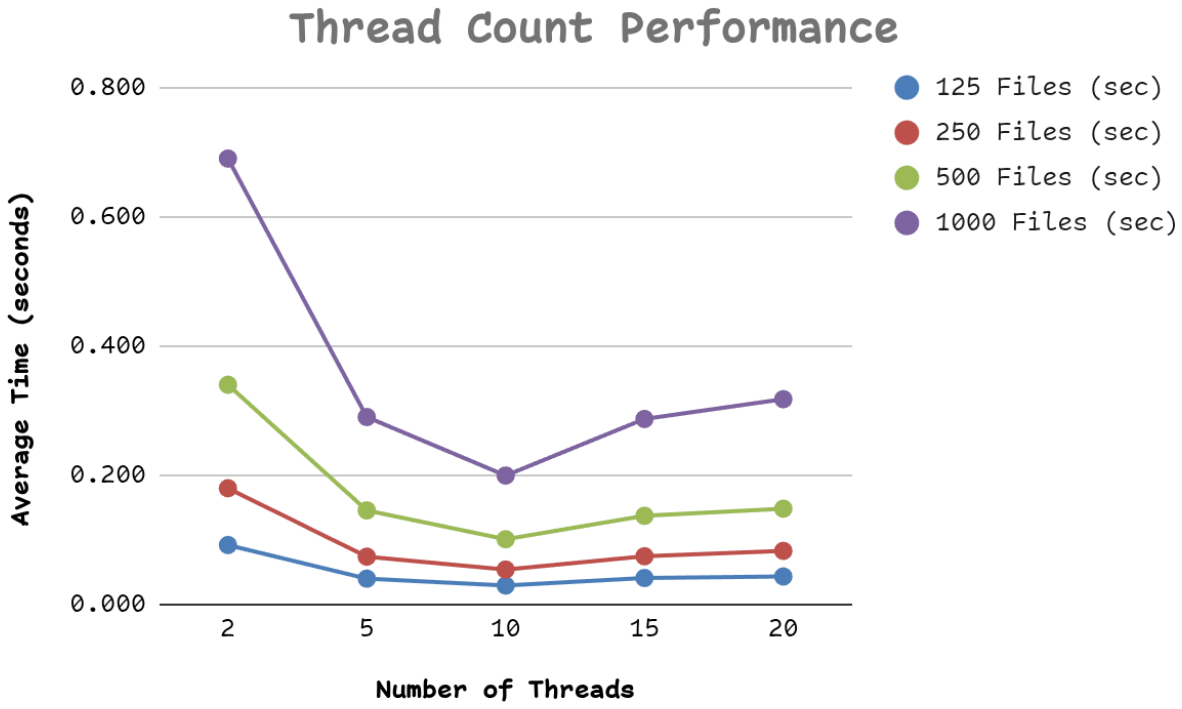
How does the search time increase with the number of files. Does the data make sense?

Answer:

The time grows in a straight line with the number of files. Look at the numbers: 125 files took ~0.28s, 250 files took ~0.50s, 500 files took ~0.99s, 1000 files took ~2.00s. Each time I double the files, the time doubles too.

This makes perfect sense. With one thread, the program reads files one after another. If reading one file takes a tiny bit of time, then reading twice as many files takes twice as much time. There is no shortcut.

Experiment 2:



Does increasing the thread count always improve performance? If your data shows otherwise (i.e., as the number of threads keeps increasing, at some point performance decreases), then what might be a possible reason?

Answer:

No. The data clearly shows there is a sweet spot, and after that, more threads actually make things ***slower***. For 1000 files: 2 threads -> 0.69s, 5 threads -> 0.29s, 10 threads -> 0.20s (fastest), 15 threads -> 0.29s, 20 threads -> 0.32s.

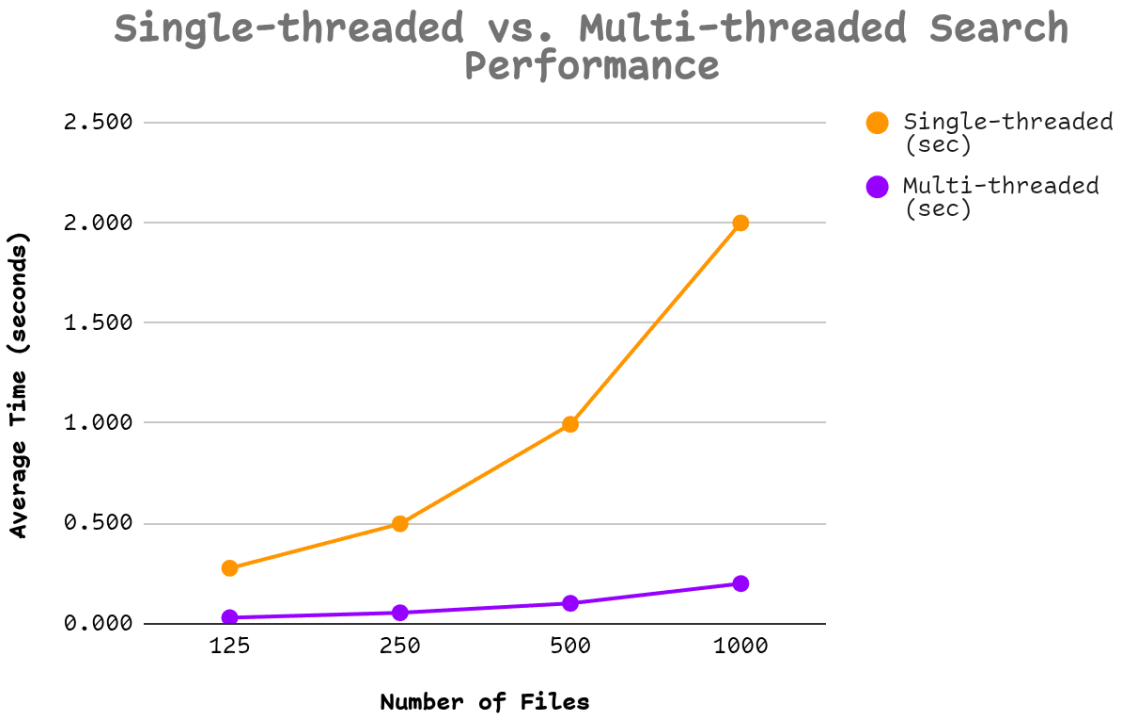
The pattern: speed improves up to 10 threads, then gets worse. Three likely reasons:

1. Too many threads, not enough CPU cores. My computer has a fixed number of cores (probably 8-12). If I create 20 threads but only have 10 cores, the operating system has to keep pausing one thread to let another run. This pausing and resuming (context switching) wastes time.

2. Overhead of creating and managing threads. Each new thread costs something to set up, coordinate, and clean up. With 20 threads splitting 125 files, each thread only handles ~6 files the setup cost starts to outweigh the actual work.

3. Threads fighting over shared resources. Threads share the disk, the cache, and any locks More threads means more waiting in line for these shared resources.

Experiment 3:



Does the concurrency brough about by multi-threading improves search time when compared with single-threaded performance? Why multiple threads are faster than a single one?

Answers:

Yes, by a huge margin, about 10x faster across every file size. The single-threaded program took 2.00s on 1000 files; the 10-thread version did it in 0.20s.

Why multiple threads win: Modern CPUs contain multiple cores, so multiple threads can execute truly in parallel. A single thread can typically utilize only one core, leaving the others underused. When searching many files for a word, the work is highly parallelizable because each file can be scanned independently. As a result, using multiple threads can significantly reduce runtime, often approaching linear scaling with the number of cores. However, perfect speedup is uncommon due to thread overhead, I/O limits, memory bandwidth, load imbalance, and the small serial portions of the program.